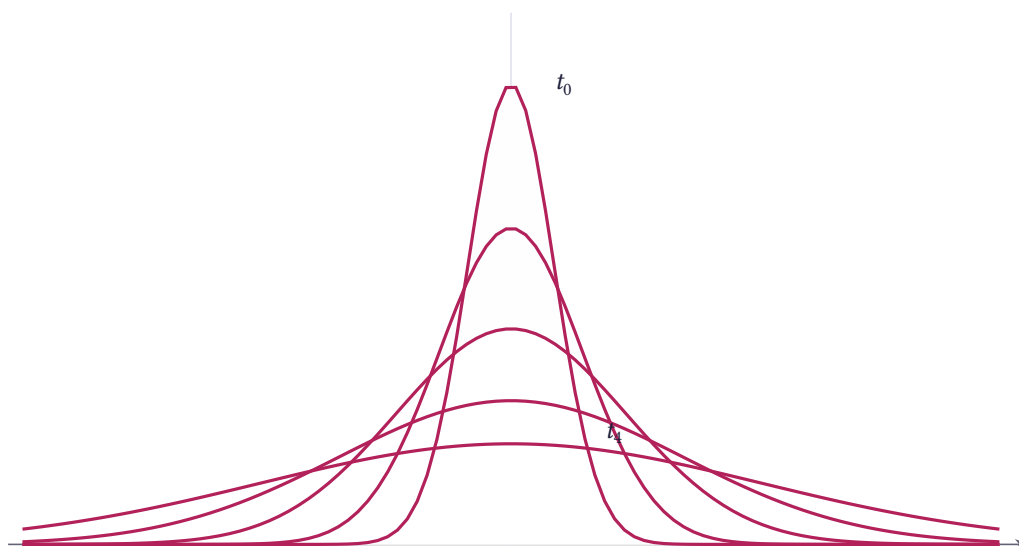


d'Alembert's Method and the One-Dimensional Heat Equation

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12.1 d'Alembert's Method for the One-Dimensional Wave Equation

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} \\ u(0, t) = u(L, t) = 0 \\ u(x, 0) = f(x) \\ u_t(x, 0) = g(x) \end{cases}$$

We rewrite and solve the equation using the following change of variables.

$$u(x, t) \Rightarrow u(\varphi, \psi), \quad \begin{cases} \varphi = x + ct \\ \psi = x - ct \end{cases}$$

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial \varphi} \frac{\partial \varphi}{\partial x} + \frac{\partial u}{\partial \psi} \frac{\partial \psi}{\partial x} = \frac{\partial u}{\partial \varphi} + \frac{\partial u}{\partial \psi}$$

$$\frac{\partial u}{\partial t} = \frac{\partial u}{\partial \varphi} \frac{\partial \varphi}{\partial t} + \frac{\partial u}{\partial \psi} \frac{\partial \psi}{\partial t} = c \frac{\partial u}{\partial \varphi} - c \frac{\partial u}{\partial \psi}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial \varphi} + \frac{\partial u}{\partial \psi} \right) = \left(\frac{\partial^2 u}{\partial \varphi^2} + \frac{\partial^2 u}{\partial \varphi \partial \psi} \right) \cdot 1 + \left(\frac{\partial^2 u}{\partial \psi \partial \varphi} + \frac{\partial^2 u}{\partial \psi^2} \right) \cdot 1 = \frac{\partial^2 u}{\partial \varphi^2} + 2 \frac{\partial^2 u}{\partial \varphi \partial \psi} + \frac{\partial^2 u}{\partial \psi^2}$$

(using $\frac{\partial^2 u}{\partial \varphi \partial \psi} = \frac{\partial^2 u}{\partial \psi \partial \varphi}$.)

Exercise 12.1

In the same way, prove the identity

$$\frac{\partial^2 u}{\partial t^2} = c^2 \left(\frac{\partial^2 u}{\partial \varphi^2} - 2 \frac{\partial^2 u}{\partial \varphi \partial \psi} + \frac{\partial^2 u}{\partial \psi^2} \right)$$

Substituting the two relations above into the one-dimensional wave equation:

$$c^2 \left(\frac{\partial^2 u}{\partial \varphi^2} - 2 \frac{\partial^2 u}{\partial \varphi \partial \psi} + \frac{\partial^2 u}{\partial \psi^2} \right) = c^2 \left(\frac{\partial^2 u}{\partial \varphi^2} + 2 \frac{\partial^2 u}{\partial \varphi \partial \psi} + \frac{\partial^2 u}{\partial \psi^2} \right) \implies \frac{\partial^2 u}{\partial \varphi \partial \psi} = 0$$

The general solution:

$$\frac{\partial u}{\partial \varphi} = \phi_1(\varphi) \implies u(\varphi, \psi) = \int \phi_1(\varphi) d\varphi + \Psi(\psi) = \Phi(\varphi) + \Psi(\psi) \implies u(x, t) = \Phi(x + ct) + \Psi(x - ct)$$

Applying the initial conditions:

$$u(x, 0) = \Phi(x) + \Psi(x) = f(x)$$

$$u_t(x, t) = \frac{\partial \Phi}{\partial \varphi} \frac{\partial \varphi}{\partial t} + \frac{\partial \Psi}{\partial \psi} \frac{\partial \psi}{\partial t} = c\Phi'(\varphi) - c\Psi'(\psi) = c\Phi'(x + ct) - c\Psi'(x - ct)$$

$$u_t(x, 0) = c\Phi'(x) - c\Psi'(x) = c[\Phi'(x) - \Psi'(x)] = g(x)$$

$$\begin{cases} \Phi(x) + \Psi(x) = f(x) \\ \Phi'(x) - \Psi'(x) = \frac{1}{c}g(x) \end{cases} \implies \begin{cases} \Phi'(x) + \Psi'(x) = f'(x) \\ \Phi'(x) - \Psi'(x) = \frac{1}{c}g(x) \end{cases} \implies \begin{cases} \Phi'(x) = \frac{1}{2}f'(x) + \frac{1}{2c}g(x) \\ \Psi'(x) = \frac{1}{2}f'(x) - \frac{1}{2c}g(x) \end{cases}$$

$$\begin{cases} \Phi(x) = \frac{1}{2}f(x) + \frac{1}{2c} \int_0^x g(s) ds + k_1 \\ \Psi(x) = \frac{1}{2}f(x) - \frac{1}{2c} \int_0^x g(s) ds + k_2 \end{cases}$$

$$u(x, t) = \frac{1}{2} \left(f(x+ct) + \frac{1}{c} \int_0^{x+ct} g(s) ds \right) + \frac{1}{2} \left(f(x-ct) - \frac{1}{c} \int_0^{x-ct} g(s) ds \right) + k_1 + k_2$$

Key result 12.2 ■ d'Alembert's formula

$$u(x, t) = \frac{1}{2} \left[f(x+ct) + f(x-ct) \right] + \frac{1}{2c} \int_{x-ct}^{x+ct} g(s) ds + K, \quad u(0, 0) = 0 \Rightarrow K = 0$$

$$u(x, t) = \frac{1}{2} \left[f(x+ct) + f(x-ct) \right] + \frac{1}{2c} \int_{x-ct}^{x+ct} g(s) ds$$

Exercise 12.3

Prove that if the boundary conditions are to hold, the functions $f(x)$ and $g(x)$ in the formula above must both be periodic with period $2L$ and odd.

12.2 The Heat Equation

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}, \quad \frac{\partial u}{\partial t} = c^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right), \quad \frac{\partial u}{\partial t} = c^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right)$$

■ The One-Dimensional Heat Equation

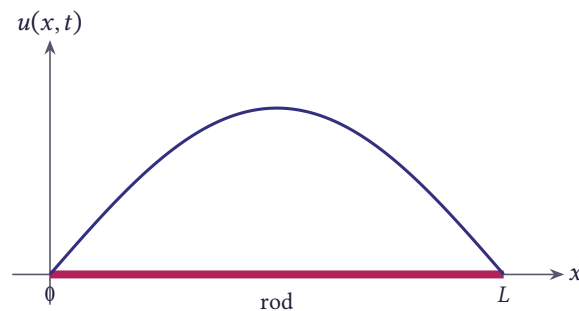


Figure 12.1 A one-dimensional rod $0 \leq x \leq L$ with temperature distribution $u(x, t)$

For a unique solution we need boundary conditions and initial conditions:

- Boundary conditions: $u(0, t) = u(L, t) = 0$
- Initial conditions: $u(x, 0) = f(x)$

12.3 Solving the One-Dimensional Heat Equation

$$\begin{cases} \frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2} \\ u(0, t) = u(L, t) = 0 \\ u(x, 0) = f(x) \end{cases}$$

Here we solve the equation by **separation of variables**. Assume $u(x, t) = X(x) \times T(t)$.

$$\frac{\partial^2 u}{\partial x^2} = X''(x)T(t), \quad \frac{\partial u}{\partial t} = X(x)T'(t) \implies X(x)T'(t) = c^2 X''(x)T(t) \implies \frac{T'}{c^2 T} = \frac{X''}{X} = k$$

k must necessarily be negative (for $k = 0$ and $k > 0$ one gets $X(x) = 0$ exactly as in the wave problem).

With $k = -\lambda^2$:

$$\frac{T'}{c^2 T} = \frac{X''}{X} = -\lambda^2 \implies \begin{cases} X'' + \lambda^2 X = 0 \\ T' + c^2 \lambda^2 T = 0 \end{cases}$$

$$X(x) = A \cos \lambda x + B \sin \lambda x, \quad X(0) = 0 \implies A = 0, \quad X(L) = 0 \implies B \sin \lambda L = 0 \implies \lambda = \frac{n\pi}{L} = \lambda_n$$

$$X_n(x) = B_n \sin \lambda_n x, \quad T' + c^2 \lambda_n^2 T = 0 \implies T_n(t) = C_n e^{-(c\lambda_n)^2 t}$$

$$u_n(x, t) = X_n(x) \times T_n(t) = B_n \sin\left(\frac{n\pi}{L}x\right) \times C_n e^{-\left(\frac{cn\pi}{L}\right)^2 t}$$

$$u(x, t) = \sum_{n=1}^{\infty} A_n e^{-(c\lambda_n)^2 t} \sin(\lambda_n x), \quad \lambda_n = \frac{n\pi}{L}$$

The initial condition $u(x, 0) = f(x)$:

$$u(x, 0) = \sum_{n=1}^{\infty} A_n \sin \lambda_n x = f(x)$$

$f(x)$ has been set equal to a periodic odd function, so to compute the unknown Fourier coefficients we must regard $f(x)$ as a periodic odd function (with period $2L$):

$$A_n = \frac{2}{2L} \int_{-L}^L f(x) \sin\left(\frac{2\pi}{2L}nx\right) dx = \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi}{L}x\right) dx = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi}{L}x\right) dx$$

Key result 12.4 ■ Solution of the one-dimensional heat equation (separation of variables)

$$u(x, t) = \sum_{n=1}^{\infty} A_n e^{-(c\lambda_n)^2 t} \sin(\lambda_n x), \quad \lambda_n = \frac{n\pi}{L}, \quad A_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi}{L}x\right) dx$$

Example 12.5 ■ Example 1

Find the solution of the following one-dimensional heat equation.

$$\begin{aligned}\frac{\partial u}{\partial t} &= \frac{\partial^2 u}{\partial x^2}, & u(0, t) = u(10, t) &= 0, & u(x, 0) &= f(x) = \begin{cases} x & 0 \leq x < 5 \\ 10 - x & 5 \leq x \leq 10 \end{cases} \\ A_n &= \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi}{L}x\right) dx = 0.2 \int_0^5 x \sin\left(\frac{n\pi}{10}x\right) dx + 0.2 \int_5^{10} (10 - x) \sin\left(\frac{n\pi}{10}x\right) dx \\ &= 0.2 \left[(x) \left(\frac{-10}{n\pi} \cos \frac{n\pi}{10}x \right) - (1) \left(\frac{-100}{n^2\pi^2} \sin \frac{n\pi}{10}x \right) \right]_0^5 \\ &\quad + 0.2 \left[(10 - x) \left(\frac{-10}{n\pi} \cos \frac{n\pi}{10}x \right) - (-1) \left(\frac{-100}{n^2\pi^2} \sin \frac{n\pi}{10}x \right) \right]_5^{10} = \frac{40}{n^2\pi^2} \sin \frac{n\pi}{2} \\ u(x, t) &= \sum_{n=1}^{\infty} \left(\frac{40}{n^2\pi^2} \sin \frac{n\pi}{2} \right) e^{-\left(\frac{cn\pi}{10}\right)^2 t} \sin\left(\frac{n\pi}{10}x\right)\end{aligned}$$

(with $c = 1$.)

Example 12.6 ■ Example 2 (nonzero boundary values)

Find the solution of the following one-dimensional heat equation.

$$\begin{aligned}\frac{\partial u}{\partial t} &= c^2 \frac{\partial^2 u}{\partial x^2}, & u(0, t) &= T_0, & u(L, t) &= T_1, & u(x, 0) &= f(x) \\ u(x, t) &= w(x, t) + v(x) \implies \begin{cases} w(0, t) + v(0) = T_0 \\ w(L, t) + v(L) = T_1 \end{cases} \\ \frac{\partial w}{\partial t} &= c^2 \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 v}{\partial x^2} \right) \\ \begin{cases} \frac{\partial^2 v}{\partial x^2} = 0 \\ v(0) = T_0 \\ v(L) = T_1 \end{cases} &\implies v(x) = \left(\frac{T_1 - T_0}{L} \right) x + T_0, & \begin{cases} \frac{\partial w}{\partial t} = c^2 \frac{\partial^2 w}{\partial x^2} \\ w(0, t) = w(L, t) = 0 \end{cases} &\implies w(x, t) \\ &= \sum_{n=1}^{\infty} A_n e^{-(c\lambda_n)^2 t} \sin(\lambda_n x) \\ u(x, t) &= \underbrace{\sum_{n=1}^{\infty} A_n e^{-(c\lambda_n)^2 t} \sin(\lambda_n x)}_{w(x, t)} + \underbrace{\left(\frac{T_1 - T_0}{L} \right) x + T_0}_{v(x)}, & \lambda_n &= \frac{n\pi}{L}\end{aligned}$$

The initial condition:

$$\begin{aligned}\sum_{n=1}^{\infty} A_n \sin \frac{n\pi}{L} x &= f(x) - \left(\frac{T_1 - T_0}{L} \right) x - T_0 \triangleq f_1(x) \implies A_n \\ &= \frac{2}{L} \int_0^L \left[f(x) - \left(\frac{T_1 - T_0}{L} \right) x - T_0 \right] \sin \frac{n\pi}{L} x dx\end{aligned}$$

Example 12.7 ■ Example 3 (insulated ends)

Find the solution of the following one-dimensional heat equation.

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}, \quad u_x(0, t) = u_x(L, t) = 0, \quad u(x, 0) = f(x)$$

We solve this problem by **separation of variables**: $u(x, t) = X(x) \times T(t)$.

$$\frac{T'}{c^2 T} = \frac{X''}{X} = -\lambda^2 \Rightarrow \begin{cases} X'' + \lambda^2 X = 0 \Rightarrow X(x) = A \cos \lambda x + B \sin \lambda x \\ T' + c^2 \lambda^2 T = 0 \end{cases}$$

$$X'(x) = -A\lambda \sin \lambda x + B\lambda \cos \lambda x, \quad X'(0) = 0 \Rightarrow B = 0, \quad X'(L) = 0 \Rightarrow -A\lambda \sin \lambda L = 0 \\ \Rightarrow \lambda L = n\pi \Rightarrow \lambda_n = \frac{n\pi}{L}$$

$$X_n(x) = A_n \cos \lambda_n x, \quad T_n(t) = C_n e^{-(c\lambda_n)^2 t}$$

$$u(x, t) = \sum_{n=1}^{\infty} B_n e^{-(c\lambda_n)^2 t} \cos(\lambda_n x), \quad \lambda_n = \frac{n\pi}{L}$$

The initial condition $u(x, 0) = \sum_{n=1}^{\infty} B_n \cos\left(\frac{n\pi}{L}x\right) = f(x)$ gives

$$B_n = \frac{2}{L} \int_0^L f(x) \cos\left(\frac{n\pi}{L}x\right) dx$$

12.4 The One-Dimensional Heat Equation on an Infinite Rod

$$\begin{cases} \frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2} \\ u(\infty, t) = u(-\infty, t) = 0 \\ u(x, 0) = f(x), \quad -\infty < x < \infty \end{cases}$$

Assuming $u(x, t) = X(x) \times T(t)$:

$$X(x)T'(t) = c^2 X''(x)T(t) \Rightarrow \frac{T'}{c^2 T} = \frac{X''}{X} = -\lambda^2 \\ \Rightarrow \begin{cases} X(x) = A_1(\lambda) \cos \lambda x + B_1(\lambda) \sin \lambda x = X_\lambda(x) \\ T(t) = C_1(\lambda) e^{-c^2 \lambda^2 t} = T_\lambda(t) \end{cases}$$

$$u_\lambda(x, t) = [A(\lambda) \cos \lambda x + B(\lambda) \sin \lambda x] e^{-c^2 \lambda^2 t} \implies u(x, t) \\ = \frac{1}{\pi} \int_0^\infty [A(\lambda) \cos \lambda x + B(\lambda) \sin \lambda x] e^{-c^2 \lambda^2 t} d\lambda$$

Checking the equation:

$$\frac{\partial^2 u}{\partial x^2} = \frac{1}{\pi} \int_0^\infty -\lambda^2 [A \cos \lambda x + B \sin \lambda x] e^{-c^2 \lambda^2 t} d\lambda, \quad \frac{\partial u}{\partial t} \\ = \frac{1}{\pi} \int_0^\infty -c^2 \lambda^2 [A \cos \lambda x + B \sin \lambda x] e^{-c^2 \lambda^2 t} d\lambda \Rightarrow \frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2} \checkmark$$

Applying the initial condition: $u(x, 0) = \frac{1}{\pi} \int_0^\infty [A(\lambda) \cos \lambda x + B(\lambda) \sin \lambda x] d\lambda = f(x)$, i.e. the Fourier integral (Chapter 10):

$$\begin{aligned} A(\lambda) &= \mathfrak{F}_C\{f(x)\} = \int_{-\infty}^\infty f(x) \cos \lambda x dx, & B(\lambda) &= \mathfrak{F}_S\{f(x)\} = \int_{-\infty}^\infty f(x) \sin \lambda x dx \\ u(x, t) &= \frac{1}{\pi} \int_0^\infty \left[\left(\int_{-\infty}^\infty f(v) \cos \lambda v dv \right) \cos \lambda x + \left(\int_{-\infty}^\infty f(v) \sin \lambda v dv \right) \sin \lambda x \right] e^{-c^2 \lambda^2 t} d\lambda \\ &= \frac{1}{\pi} \int_0^\infty \left\{ \int_{-\infty}^\infty f(v) [\cos \lambda v \cos \lambda x + \sin \lambda v \sin \lambda x] dv \right\} e^{-c^2 \lambda^2 t} d\lambda \\ &= \frac{1}{\pi} \int_0^\infty \left\{ \int_{-\infty}^\infty f(v) \cos \lambda(x-v) dv \right\} e^{-c^2 \lambda^2 t} d\lambda \end{aligned}$$

Key result 12.8 ■ First form of the solution

$$u(x, t) = \frac{1}{\pi} \int_{-\infty}^\infty f(v) \left\{ \int_0^\infty e^{-c^2 \lambda^2 t} \cos \lambda(x-v) d\lambda \right\} dv$$

Mathematical Lemmas

Theorem 12.9 ■ Lemma (a)

$$I = \int_0^\infty e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$$

Proof.

$$\begin{aligned} I^2 &= \int_0^\infty e^{-x^2} dx \times \int_0^\infty e^{-y^2} dy = \int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dx dy \stackrel{\rho^2=x^2+y^2}{=} \int_0^{\pi/2} \int_0^\infty \rho e^{-\rho^2} d\rho d\theta \\ &= \int_0^{\pi/2} \left[\frac{-1}{2} e^{-\rho^2} \right]_0^\infty d\theta = \frac{\pi}{4} \end{aligned}$$

■

Theorem 12.10 ■ Lemma (b)

$$\int_0^\infty e^{-s^2} \cos(2bs) ds = \frac{\sqrt{\pi}}{2} e^{-b^2}$$

Proof.

$$I(b) = \int_0^\infty e^{-s^2} \cos(2bs) ds \Rightarrow \frac{dI(b)}{db} = - \int_0^\infty 2se^{-s^2} \sin(2bs) ds$$

Integrating by parts ($du = -2se^{-s^2} ds$, $v = \sin 2bs$):

$$\begin{aligned} \frac{dI(b)}{db} &= e^{-s^2} \sin 2bs \Big|_0^\infty - 2b \int_0^\infty e^{-s^2} \cos 2bs ds = -2b I(b) \Rightarrow I(b) = ke^{-b^2}, \quad I(0) = k = \int_0^\infty e^{-s^2} ds \\ &= \frac{\sqrt{\pi}}{2} \end{aligned}$$

With the substitution $c\lambda\sqrt{t} = s$:

$$\int_0^\infty e^{-c^2\lambda^2 t} \cos \lambda(x-v) d\lambda = \frac{1}{c\sqrt{t}} \int_0^\infty e^{-s^2} \cos \left[2 \frac{(x-v)}{2c\sqrt{t}} s \right] ds = \frac{1}{c\sqrt{t}} \frac{\sqrt{\pi}}{2} e^{-\left(\frac{x-v}{2c\sqrt{t}}\right)^2}$$

Key result 12.11 ■ Second form of the solution

$$u(x, t) = \frac{1}{2c\sqrt{\pi t}} \int_{-\infty}^\infty f(v) e^{-\frac{(x-v)^2}{4c^2 t}} dv$$

With the substitution $z = \frac{v-x}{2c\sqrt{t}}$ (i.e. $v = x + 2c\sqrt{t} z$ and $dv = 2c\sqrt{t} dz$):

Key result 12.12 ■ Third form of the solution

$$u(x, t) = \frac{1}{\sqrt{\pi}} \int_{-\infty}^\infty f(x + 2c\sqrt{t} z) e^{-z^2} dz$$

12.5 The Error Function

In mathematics the error function is most often defined as

$$\operatorname{erf}(x) \triangleq \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$$

Properties of the error function:

- odd symmetry: $\operatorname{erf}(-x) = -\operatorname{erf}(x)$
- asymptotic property: $\operatorname{erf}(\infty) = 1$
- its Maclaurin expansion is given below (start from the expansion of e^x and prove this identity):

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \left[x - \frac{x^3}{1! \times 3} + \frac{x^5}{2! \times 5} - \frac{x^7}{3! \times 7} + \dots \right] = \frac{2}{\sqrt{\pi}} \sum_{n=1}^\infty (-1)^{n+1} \frac{x^{2n-1}}{(n-1)!(2n-1)}$$

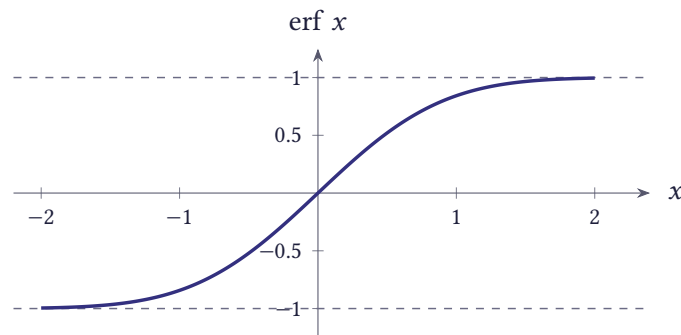


Figure 12.2 The graph of the error function $\operatorname{erf}(x)$ for $-2 \leq x \leq 2$

Example 12.13 ■ Example 1

Find the solution of the equation above for the following initial condition.

$$f(x) = \begin{cases} U_0 & |x| < 1 \\ 0 & |x| \geq 1 \end{cases}$$

We use the second form of the solution:

$$\begin{aligned} u(x, t) &= \frac{U_0}{2c\sqrt{\pi t}} \int_{-1}^1 e^{-\frac{(x-v)^2}{4c^2 t}} dv \stackrel{z=\frac{v-x}{2c\sqrt{t}}}{=} \frac{U_0}{\sqrt{\pi}} \int_{-\frac{1+x}{2c\sqrt{t}}}^{\frac{1-x}{2c\sqrt{t}}} e^{-z^2} dz \\ &= \frac{U_0}{\sqrt{\pi}} \left\{ \int_{-\frac{1+x}{2c\sqrt{t}}}^0 e^{-z^2} dz + \int_0^{\frac{1-x}{2c\sqrt{t}}} e^{-z^2} dz \right\} = \frac{U_0}{\sqrt{\pi}} \frac{\sqrt{\pi}}{2} \left\{ \frac{2}{\sqrt{\pi}} \int_0^{\frac{1+x}{2c\sqrt{t}}} e^{-z^2} dz + \frac{2}{\sqrt{\pi}} \int_0^{\frac{1-x}{2c\sqrt{t}}} e^{-z^2} dz \right\} \\ u(x, t) &= \frac{U_0}{2} \left[\operatorname{erf}\left(\frac{1+x}{2c\sqrt{t}}\right) + \operatorname{erf}\left(\frac{1-x}{2c\sqrt{t}}\right) \right] \end{aligned}$$

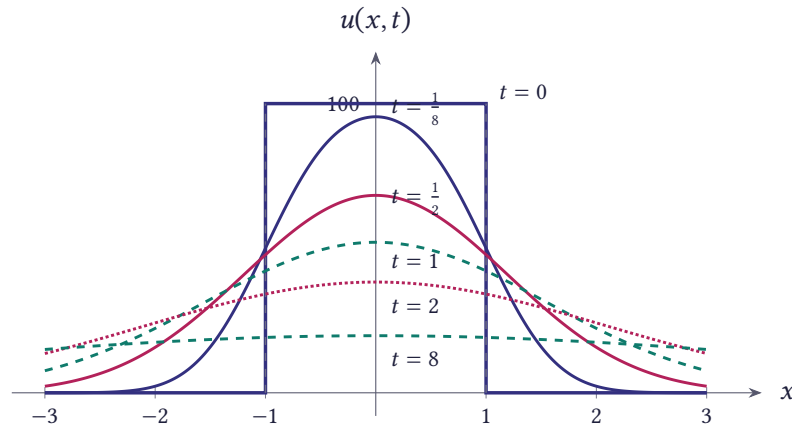


Figure 12.3 The temperature distribution $u(x, t)$ at $t = 0, \frac{1}{8}, \frac{1}{2}, 1, 2$ and 8 for the initial pulse $U_0 = 100$ and $c = 1$

Example 12.14 ■ Example 2

Find the solution of the equation above for the following initial condition.

$$f(x) = \begin{cases} 1 & x > 0 \\ 0 & x \leq 0 \end{cases}$$

We use the second form of the solution:

$$\begin{aligned} u(x, t) &= \frac{1}{2c\sqrt{\pi t}} \int_0^\infty e^{-\frac{(x-v)^2}{4c^2 t}} dv \stackrel{z=\frac{v-x}{2c\sqrt{t}}}{=} \frac{1}{\sqrt{\pi}} \int_{-\frac{x}{2c\sqrt{t}}}^\infty e^{-z^2} dz = \frac{1}{\sqrt{\pi}} \left\{ \int_{-\frac{x}{2c\sqrt{t}}}^0 e^{-z^2} dz + \int_0^\infty e^{-z^2} dz \right\} \\ u(x, t) &= \frac{1}{2} \left[\operatorname{erf}\left(\frac{x}{2c\sqrt{t}}\right) + 1 \right] \end{aligned}$$